A SHORTER PRODUCT RELATION FOR OPENAI'S MULTIPLICITY-TWO/THREE ZARISKI COUNTEREXAMPLE

[AUTHOR NAME]

Abstract. OpenAI's preprint *Ambiently homeomorphic isolated hypersurfaces of multiplicities two and three* (`openai/math`, family 059) answers Zariski's embedded multiplicity question negatively by a pair of weighted homogeneous germs in $\mathbb{C}^N$, where $N = 4s_0 + 16$ and s_0 is determined by an integer relation among the rational numbers $A_b = (d-b)/b$ and $R_b = (2d+b)(2d-b)/((3d-b)(d+b))$, $d = 3^{11}$. The relation supplied with the preprint has 140-digit coefficients and gives $N \approx 4.58 \cdot 10^{143}$. We exhibit a relation with all $|e_b| \le 29$ and $\sum e_b = 206$; the unchanged construction then gives $N = 1{,}802{,}976$. The argument is OpenAI's; we only find a shorter relation, and we assume the preprint is correct.

1. The finite input

Let $d = 3^{11}$, $D = 3d$, and $\mathcal{B} = \{b \in \mathbb{Z} : 0 < b < d/2,\ 3 \nmid b\}$.

Lemma 1 ([1, Lemma `seed:product`])**.** *There are integers $(e_b)_{b\in\mathcal{B}}$ with $E := \sum_b e_b > 0$ and $\prod_b A_b^{e_b} = \prod_b R_b^{e_b} = 1$.*

The preprint proves Lemma 1 by a rank certificate over $\mathbb{F}_{1009}$ and the adjugate of an 868×868 minor, and supplies, as an independent witness, an explicit relation with 849 nonzero coefficients of up to 140 decimal digits [1, App. A].

From any such relation the preprint builds ([1, Prop. `seed:seeds`]) two sums of chain polynomials p_1, p_2 in s_0 variables. For each b, the four starting weights $3b/D$, b/D, $(d-b)/D$, $(d+b)/D$ occur with signed multiplicities $2e_b, -2e_b, 2e_b, 2e_b$ (positive means side 2). Each occurrence is completed to an isolated chain by the decreasing-remainder algorithm, and its tail is copied to the opposite side. Side 1 receives $4E$ squares and both sides receive two cubes. Writing $\ell(c)$ for the length of the chain that starts at c/D, one obtains

$$s_0 = \sum_b |e_b|\, c_b + 2E + 2, \qquad c_b = 4 + 2 \sum_{c\in\{3b,b,d-b,d+b\}} (\ell(c) - 1), \qquad N = 4s_0 + 16. \tag{1}$$

2. What the proof uses from the relation

We read the full preprint and list every use of (e_b):

1. $e_b \in \mathbb{Z}$ and $b \in \mathcal{B}$: the weight table and chain completion (`seeds.tex`:131–154);
2. $E > 0$: the squares sit on side 1, so $\operatorname{ord}_0 p_1 = 2$, and f_1 has Hessian rank $16E > 0$ (`seeds.tex`:156–166; `conclusion.tex`:284–287, 391–394);
3. parity: all signed counts are even, so s_0 is even (`seeds.tex`:157–161);
4. $\prod A_b^{2e_b} = 1$ and $\prod (R_b/A_b)^{2e_b} = 1$: Case 1 of the mod-2 spectral comparison (`seeds.tex`:197–212).

Case 2 (the triple-angle identity) holds row by row for any e_b. Case 3, the monodromy order $D = 3^{12}$, and everything in Sections 4–6 of the preprint use only the conclusions of Corollary `seed:prepared`, which depend on (1)–(4). The adjugate form of the witness is used only for existence and finiteness (`seeds.tex`:58–76).

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Theorem 2. *The integers e_b in the file* `relation_d3pow11_N1802976.json` *(*1160 *nonzero entries,* $\max|e_b| = 29$*,* $E = 206$*,* $\sum|e_b| = 7414$*) satisfy Lemma 1. Consequently, if the preprint's proof is correct, the conclusion of its Theorem 1.1 (reduced, isolated, weighted homogeneous, real-coefficient germs f_1, f_2 with* $\operatorname{ord}_0 f_1 = 2$*,* $\operatorname{ord}_0 f_2 = 3$*, ambiently homeomorphic) holds in $\mathbb{C}^N$ with*

$$s_0 = 450{,}740, \qquad N = 1{,}802{,}976 = 8 \cdot 225{,}372,$$

and the Hessian coranks are $N - 3296$ and N.

Proof. The relation is verified by exact integer arithmetic in two ways. (a) The six integers of each row are factored, using the preprint's own verifier and, separately, our own sieve-based code. (b) Both products are formed directly as big integers, without factoring: numerator and denominator agree, with 118,503 and 272,384 bits respectively. Inserting the relation in (1) gives $s_0 = 450{,}740$. The rest follows from the preprint, since by §2 it uses nothing beyond properties (1)–(4). □

Remark 3 (Additional check). Equality of the two spectra modulo 2 is equivalent to the equality of the two Fourier transforms $P_{p_j}(e^{\pi i k \cdot})$ for all k modulo $4D$. We assembled both full weight multisets explicitly (450,738 weights per side before the cubes) and compared the transforms at 398 characters (all $k < 60$ and 400 random k; k odd with $3 \mid k$ is trivial). The angles were reduced exactly before floating-point evaluation. The maximal discrepancy is 10^{-9}. Perturbing one coefficient, or flipping one sign, gives discrepancies ≥ 0.57.

3. How the relation was found

Keep the columns $b \in \mathcal{B}$ for which $d - b, b, 2d \pm b, 3d - b, d + b$ are 6000-smooth. Delete singly-supported columns repeatedly, as in [1, proof of Lemma `seed:product`]. This leaves 1405 columns and 1150 nonzero valuation rows. The $\mathbb{Q}$-kernel has dimension 255; saturate it to the integer kernel lattice. Apply LLL after scaling column b by c_b (cf. (1)). Among the reduced vectors and their pairwise sums and differences, choose the one with $E \neq 0$ minimizing (1), and change its sign if needed. This search is heuristic. Only its output is used, and that output is verified as above.

Remark 4 (Limitations). $N = 1{,}802{,}976$ is an upper bound for this particular construction. We do not claim that it is optimal. Partial runs found no relation with $E \neq 0$ for $d \leq 3^8$, nor for $d = 3^9$ with smoothness bound 10^5. These runs prove nothing about minimality. The topological and lattice-theoretic parts of [1] were not re-verified.

References

[1] OpenAI, *Ambiently homeomorphic isolated hypersurfaces of multiplicities two and three*, OpenAI Math Release preprint, September 24, 2026, `https://github.com/openai/math` (commit `fd4aeeb2`), directory `preprints/Ambiently-homeomorphic-isolated-hypersurfaces-of-multiplicities-two-and-three-September-24-2026`.

[2] A. K. Lenstra, H. W. Lenstra, L. Lovász, *Factoring polynomials with rational coefficients*, Math. Ann. 261 (1982), 515–534.

[3] O. Zariski, *Some open questions in the theory of singularities*, Bull. Amer. Math. Soc. 77 (1971), 481–491.

Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.